

Module 3: Data Link Layer and Wireless Networks



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Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

Wireless Networks

- Wireless LANs
- Mobile IP

Hands-On

- Datalink Provider Interface
- SOCK_PACKET
- PF_PACKET

Wired LANs: Ethernet Protocol

- TCP/IP does not define protocols for the Data-Link and Physical layers.
- These layers are implemented by network technologies such as LANs and WANs.
- Networks may be wired or wireless.
- This section focuses on wired LANs, specifically Ethernet.

Observation

Ethernet is the dominant wired LAN technology used today.

Local Area Networks (LANs)

- A LAN is a computer network designed for a limited geographic area.
- Examples:
 - Building
 - Office
 - School
 - University Campus
- LANs enable resource sharing among connected devices.
- Modern LANs are usually connected to the Internet through routers.



Evolution of LAN Technologies

During the 1980s and 1990s, several LAN technologies were used:

- Ethernet
- Token Ring
- Token Bus
- FDDI (Fiber Distributed Data Interface)
- ATM LAN



Media Access Methods

- Ethernet → CSMA/CD
- Token Ring → Token Passing
- Token Bus → Token Passing
- FDDI → Token Passing

Why Did Ethernet Survive?

Most LAN technologies disappeared from the market, but Ethernet survived because:

- Simple design
- Cost-effective implementation
- Backward compatibility
- Easy upgrade path
- Ability to support higher transmission rates

Key Idea

Organizations could upgrade existing Ethernet networks instead of replacing them with a completely new technology.

Ethernet's Continuous Evolution

Ethernet evolved to satisfy increasing bandwidth demands.

- Existing infrastructure could be reused.
- Higher-speed Ethernet versions remained compatible.
- Upgrades were less expensive than switching technologies.

Result

Ethernet became the dominant wired LAN technology worldwide.

History of Ethernet

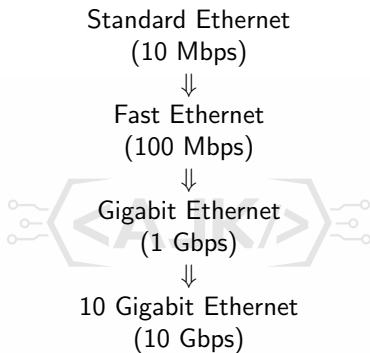
- Developed during the 1970s.
- Invented by:
 - Robert Metcalfe
 - David Boggs
- Designed as a shared-medium LAN technology.
- Initially operated at 10 Mbps.



Generations of Ethernet

Ethernet Generation	Data Rate
Standard Ethernet	10 Mbps
Fast Ethernet	100 Mbps
Gigabit Ethernet	1 Gbps
10 Gigabit Ethernet	10 Gbps

Ethernet Generations Timeline



Trend

Each new generation increased speed by approximately 10 times.

Ethernet at a Glance

Feature	Ethernet
Network Type	Wired LAN
Original Access Method	CSMA/CD
Inventors	Robert Metcalfe, David Boggs
First Generation Speed	10 Mbps
Modern Speeds	Up to 10 Gbps and beyond
Status	Dominant LAN Technology

Summary

- Ethernet is the most successful wired LAN technology.
- It uses Data-Link and Physical layer protocols outside TCP/IP.
- Competing LAN technologies gradually disappeared.
- Ethernet survived because of scalability and backward compatibility.
- Ethernet evolved through four major generations:
 - Standard Ethernet
 - Fast Ethernet
 - Gigabit Ethernet
 - 10 Gigabit Ethernet

Key Takeaway

Ethernet's ability to evolve with increasing bandwidth demands made it the universal wired LAN technology.

IEEE Project 802

- Started by the IEEE Computer Society in 1985.
- Goal: Establish standards for LAN communication.
- Ensures interoperability among equipment from different manufacturers.
- Focuses on the Physical Layer and Data-Link Layer.
- Does not replace:
 - OSI Model
 - TCP/IP Protocol Suite

Purpose

Provide standardized LAN protocols for reliable communication across different devices and vendors.

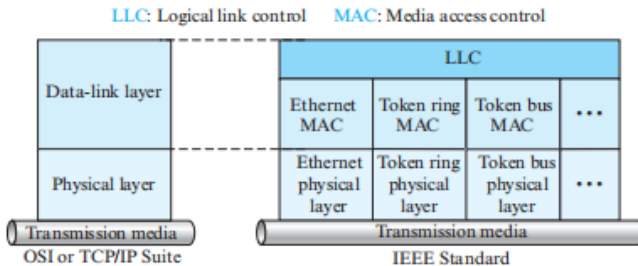
IEEE 802 and TCP/IP Relationship

- IEEE 802 specifies standards for:
 - Physical Layer
 - Data-Link Layer
- The Data-Link Layer is divided into two sublayers:
 - 1 Logical Link Control (LLC)
 - 2 Media Access Control (MAC)

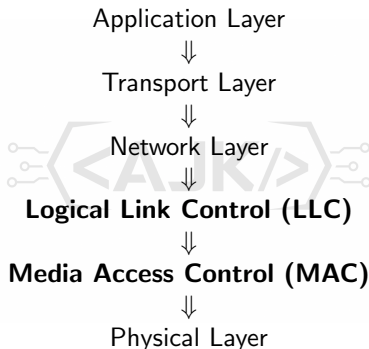
Important

IEEE 802 complements TCP/IP by defining LAN-specific functions at the lower layers.

Figure 5.54 IEEE standard for LANs



IEEE 802 Data-Link Layer Structure



Logical Link Control (LLC)

The LLC sublayer handles common data-link functions for all IEEE LANs.

Functions of LLC

- Flow Control
- Error Control
- Part of Framing
- Provides a common interface to upper layers



Advantage

LLC provides a single link-layer control protocol for all IEEE LAN technologies.

Benefits of LLC

- Makes MAC implementation transparent.
- Supports communication between different LAN technologies.
- Simplifies network-layer interaction.
- Provides uniform services regardless of underlying LAN type.

Key Point

LLC enables interoperability among different IEEE LANs.

Media Access Control (MAC)

The MAC sublayer controls how devices access the shared transmission medium.

Responsibilities

- Media Access Management
- Frame Delimiting
- Addressing Support
- Part of Framing Functions



MAC in Different LAN Technologies

LAN Technology	MAC Access Method
Ethernet	CSMA/CD
Token Ring	Token Passing
Token Bus	Token Passing
Wireless LAN	CSMA/CA

LLC vs MAC

Feature	LLC	MAC
Main Function	Link Control	Media Access
Flow Control	Yes	No
Error Control	Yes	No
Framing Support	Partial	Partial
Access Method Control	No	Yes
LAN Independence	Common to All	LAN Specific

Summary

- IEEE Project 802 was introduced in 1985.
- It standardizes Physical and Data-Link Layer functions.
- The Data-Link Layer is divided into:
 - Logical Link Control (LLC)
 - Media Access Control (MAC)
- LLC handles flow control, error control, and part of framing.
- MAC handles media access methods and part of framing.
- Different LAN technologies use different MAC mechanisms.

Key Takeaway

IEEE 802 provides a common framework that enables interoperability among different LAN technologies.

Standard Ethernet

- Original Ethernet technology with a data rate of **10 Mbps**.
- Forms the foundation for all later Ethernet generations.
- Key concepts retained in Fast Ethernet, Gigabit Ethernet, and 10-Gigabit Ethernet.
- Uses:
 - CSMA/CD access method
 - Connectionless communication
 - Unreliable service model

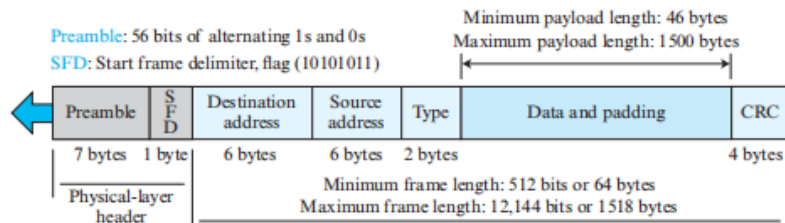
Why Study Standard Ethernet?

Understanding Standard Ethernet helps in understanding all modern Ethernet technologies.

Standard Ethernet Frame Format

Ethernet frame contains seven fields.

Figure 5.55 Ethernet frame



Preamble

- 7 bytes (56 bits)
- Alternating 0s and 1s
- Synchronizes sender and receiver clocks
- Added at Physical Layer

Start Frame Delimiter (SFD)

- 1 byte: 10101011
- Indicates start of frame
- Last two bits (11) indicate destination address follows
- Acts as frame-start flag

Address Fields and Type Field

Destination Address (DA)

- 48-bit Ethernet address
- Identifies intended receiver(s)

Source Address (SA)

- 48-bit Ethernet address
- Identifies sender



Type Field

- Identifies upper-layer protocol
- Examples:
 - IP
 - ARP
 - OSPF
- Used for multiplexing and demultiplexing

Data and CRC Fields

Data Field

- Minimum size: 46 bytes
- Maximum size: 1500 bytes
- Carries upper-layer payload
- Padding added if payload < 46 bytes

CRC Field

- Uses CRC-32
- Detects transmission errors
- Calculated over:
 - Addresses
 - Type field
 - Data field
- Corrupted frames are discarded



Connectionless Service

Ethernet provides a **connectionless service**.

- No connection establishment
- No connection termination
- Frames are independent
- Sender transmits whenever data is available
- Receiver may or may not be ready
- Lost frames are not reported

Important

Ethernet does not guarantee frame delivery.

Unreliable Service

Ethernet provides an **unreliable service**.

- CRC detects corrupted frames.
- Corrupted frames are discarded silently.
- No retransmission mechanism.
- Error recovery is handled by upper layers.

Examples

- UDP: Frame may be permanently lost.
- TCP: Retransmission occurs after timeout.

Ethernet Frame Length Restrictions

Parameter	Value
Minimum Frame Length	64 Bytes
Maximum Frame Length	1518 Bytes
Minimum Data Length	46 Bytes
Maximum Data Length	1500 Bytes

Why Minimum Frame Length is 64 Bytes?

- Required for proper operation of CSMA/CD.
- Sender must detect collision before frame transmission completes.
- Minimum frame size:

$$512 \text{ bits} = 64 \text{ bytes}$$

- Header + Trailer = 18 bytes

- Minimum payload:

$$64 - 18 = 46 \text{ bytes}$$

Reason

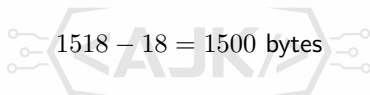
Ensures collision detection before sender finishes transmission.

Why Maximum Frame Length is 1518 Bytes?

Maximum Ethernet frame size:

1518 bytes

Payload size:


$$1518 - 18 = 1500 \text{ bytes}$$

Reasons:

- Memory was expensive when Ethernet was designed.
- Prevents one station from monopolizing the medium.
- Allows fair access among all stations.

Ethernet Addressing

- Every Ethernet device has a unique MAC address.
- Stored in the Network Interface Card (NIC).
- Length:

- Usually written in hexadecimal.

Example:

4A:30:10:21:10:1A

Transmission of Ethernet Address Bits

Addresses are transmitted:

- Left to right (byte by byte)
- Within each byte:
 - Least Significant Bit first
 - Most Significant Bit last

Advantage

The unicast/multicast bit arrives first, allowing immediate address classification.

Example 5.18: Address Transmission

Address:

47 : 20 : 1B : 2E : 08 : EE

Solution

The address is sent left to right, byte by byte; for each byte, it is sent right to left, bit by bit, as shown below:

Hexadecimal	47	20	1B	2E	08	EE
Binarys	01000111	00100000	00011011	00101110	00001000	11101110
Transmitted ←	11100010	00000100	11011000	01110100	00010000	01110111

Observation

Each byte is transmitted from least significant bit to most significant bit.

Unicast, Multicast and Broadcast Addresses

Unicast

- Sent to one receiver
- LSB of first byte = 0

Multicast

- Sent to a group
- LSB of first byte = 1



Broadcast

- Sent to all stations
- Address:

FF:FF:FF:FF:FF:FF

Address Type Identification

Figure 5.56 *Unicast and multicast addresses*



- Even second hexadecimal digit → Unicast
- Odd second hexadecimal digit → Multicast
- All Fs → Broadcast

Example 5.19: Address Classification

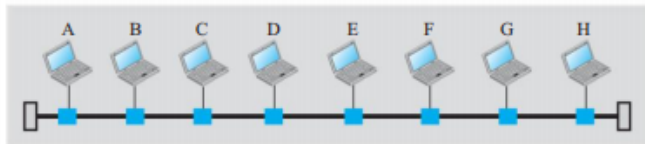
Address	Type
4A:30:10:21:10:1A	Unicast
47:20:1B:2E:08:EE	Multicast
FF:FF:FF:FF:FF:FF	Broadcast

Ethernet Transmission Types

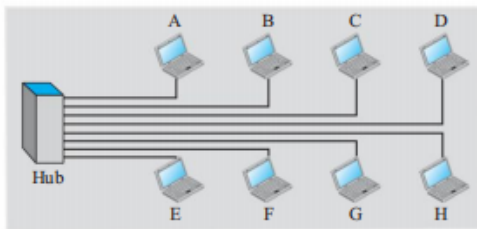
Standard Ethernet physically broadcasts every frame.

- Bus Topology: All stations receive frame.
- Star Topology: Hub forwards frame to all stations.

Figure 5.57 *Implementation of standard Ethernet*

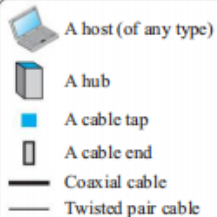


a. A LAN with a bus topology using a coaxial cable



b. A LAN with a star topology using a hub

Legend



Unicast vs Multicast vs Broadcast Transmission

Type	Frame Handling
Unicast	Only destination keeps frame
Multicast	Group members keep frame
Broadcast	All stations keep frame

Access Method: CSMA/CD

Standard Ethernet uses:

CSMA/CD + 1-Persistent Method

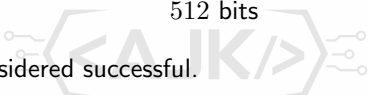
Steps:

- 1 Sense medium
- 2 If idle, transmit immediately
- 3 Monitor medium while transmitting
- 4 Detect collision if any
- 5 Send jam signal
- 6 Perform binary exponential backoff
- 7 Retransmit

Collision Detection in Ethernet

- Sender monitors medium continuously.
- If no collision detected during first:

512 bits
transmission is considered successful.



- Collision detected before 512 bits:
 - Abort transmission
 - Send jam signal
 - Execute backoff algorithm

Binary Exponential Backoff

After collision:

$$R = 0 \text{ to } (2^K - 1)$$

where:

- K = number of collisions
- Random value R selected
- Backoff time increases exponentially

Examples:

$$K = 1 \Rightarrow R = \{0, 1\}$$

$$K = 2 \Rightarrow R = \{0, 1, 2, 3\}$$

$$K = 3 \Rightarrow R = \{0, 1, 2, 3, 4, 5, 6, 7\}$$

Efficiency of Standard Ethernet

Efficiency:

$$\eta = \frac{1}{1 + 6.4a}$$

where



The diagram shows a central hexagonal node with circuit-like lines extending from its left and right sides. Inside the hexagon, the variable 'a' is defined as the ratio of Propagation Delay to Transmission Delay.

$$a = \frac{\text{Propagation Delay}}{\text{Transmission Delay}}$$

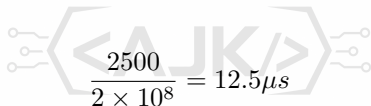
- Smaller a gives higher efficiency.
- Shorter cable increases efficiency.
- Larger frame size increases efficiency.

Example 5.20: Ethernet Efficiency

Given:

- Data Rate = 10 Mbps
- Cable Length = 2500 m
- Frame Size = 512 bits
- Propagation Speed = 2×10^8 m/s

Propagation Delay:


$$\frac{2500}{2 \times 10^8} = 12.5 \mu s$$

Transmission Delay:

$$\frac{512}{10^7} = 51.2 \mu s$$

$$a = \frac{12.5}{51.2} = 0.24$$

$$\eta = \frac{1}{1 + 6.4(0.24)} \approx 39\%$$

Standard Ethernet Implementations

Naming Convention:

10BaseX

where

- 10 = 10 Mbps
- Base = Baseband Signaling
- X = Cable Length or Medium Type

Examples:

- 10Base5
- 10Base2
- 10BaseT
- 10BaseF

Table 5.6 *Summary of Standard Ethernet implementations*

<i>Implementation</i>	<i>Medium</i>	<i>Medium Length</i>	<i>Encoding</i>
10Base5	Thick coax	500 m	Manchester
10Base2	Thin coax	185 m	Manchester
10Base-T	2 UTP	100 m	Manchester
10Base-F	2 Fiber	2000	Manchester

Summary of Standard Ethernet

- Data Rate = 10 Mbps
- Uses CSMA/CD
- Connectionless and unreliable
- Frame size:
 - Minimum = 64 bytes
 - Maximum = 1518 bytes
- Uses 48-bit MAC addresses
- Supports unicast, multicast, broadcast
- Uses binary exponential backoff
- Foundation for all modern Ethernet technologies

Fast Ethernet (100 Mbps)

- Introduced in the 1990s to compete with high-speed LAN technologies.
- Increased Ethernet speed from:

10 Mbps → 100 Mbps

- Known as **Fast Ethernet**.
- Designed to remain compatible with Standard Ethernet.

Goal

Increase transmission speed while preserving Ethernet architecture and frame format.

Need for Fast Ethernet

During the 1990s, several high-speed LAN technologies appeared:

- FDDI (Fiber Distributed Data Interface)
- Fiber Channel
- Other 100 Mbps technologies

Ethernet had to evolve to remain competitive.

Challenge

Increase speed without changing the fundamental Ethernet design.

Compatibility with Standard Ethernet

To maintain compatibility:

- MAC sublayer remained unchanged.
- Ethernet frame format remained unchanged.
- Minimum frame size remained:



- Maximum frame size remained:

1518 Bytes

Only transmission-rate dependent features required modification.

Changes Required in Fast Ethernet

Increasing speed from 10 Mbps to 100 Mbps affects:

- Access Method
- Maximum Network Length
- Physical Layer Implementation

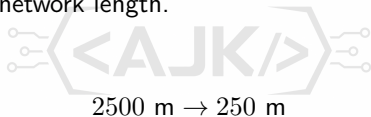
Since frames are transmitted 10 times faster:

- Collision detection must occur sooner.
- Maximum network diameter must be reduced.

Access Method: Solution 1

Solution 1: Shorter Network Length

- Retain CSMA/CD.
- Use passive hub with star topology.
- Reduce maximum network length.



Reason

Frames are transmitted 10 times faster, so collision detection must happen 10 times sooner.

Solution 2: Link-Layer Switches

- Replace shared medium with switched Ethernet.
- Use link-layer switches with buffers.
- Provide full-duplex communication.
- Create private point-to-point links.

Result

No contention and no collisions.

Switched Fast Ethernet Operation

- 1 Source host sends frame to switch.
- 2 Switch stores frame in buffer.
- 3 Switch examines destination MAC address.
- 4 Frame is forwarded through correct interface.
- 5 Destination receives frame.

Advantage

Hosts can simultaneously send and receive using full-duplex communication.

Fast Ethernet Access Methods

Approach	Description
Shared Ethernet	Uses CSMA/CD and passive hub
Switched Ethernet	Uses switches and full-duplex links
Network Length	Reduced to 250 m
Collision Domain	Eliminated in switched Ethernet

Fast Ethernet introduced:

Autonegotiation

Autonegotiation allows devices to:

- Exchange capabilities.
- Determine supported data rates.
- Select optimal operating mode.
- Ensure compatibility between devices.

Why Autonegotiation?

Different devices may support different speeds.

Example:

- Device A supports 10 Mbps only.
- Device B supports 100 Mbps and 10 Mbps.

Using autonegotiation:

100 Mbps ↔ 10 Mbps

Both devices agree to communicate at:

10 Mbps

Physical Layer Implementations

Fast Ethernet implementations are categorized as:

- 1 Two-Wire Implementations
- 2 Four-Wire Implementations

Different media types are supported:

- Shielded Twisted Pair (STP)
- Unshielded Twisted Pair (UTP)
- Fiber Optic Cable

Two-Wire Fast Ethernet

Two-wire implementations include:

100Base-TX

- Uses Shielded Twisted Pair (STP)
- 100 Mbps

100Base-FX

- Uses Fiber Optic Cable
- 100 Mbps



100Base-T4

- Uses Unshielded Twisted Pair (UTP)
- Uses four wire pairs
- Supports 100 Mbps communication

Designed specifically for UTP environments.

Fast Ethernet Implementations

Type	Medium	Speed
100Base-TX	STP	100 Mbps
100Base-FX	Fiber Optic	100 Mbps
100Base-T4	UTP	100 Mbps

Table 5.7 *Summary of Fast Ethernet implementations*

<i>Implementation</i>	<i>Medium</i>	<i>Medium Length</i>	<i>Wires</i>	<i>Encoding</i>
100Base-TX	STP	100 m	2	4B5B + MLT-3
100Base-FX	Fiber	185 m	2	4B5B + NRZ-I
100Base-T4	UTP	100 m	4	Two 8B/6T

Fast Ethernet vs Standard Ethernet

Feature	Standard Ethernet	Fast Ethernet
Speed	10 Mbps	100 Mbps
Frame Format	Same	Same
MAC Layer	Same	Same
CSMA/CD	Yes	Optional
Switch Support	Limited	Extensive
Autonegotiation	No	Yes

Summary of Fast Ethernet

- Fast Ethernet increased speed to 100 Mbps.
- Maintained compatibility with Standard Ethernet.
- MAC sublayer and frame format remained unchanged.
- Introduced autonegotiation.
- Supports:
 - 100Base-TX
 - 100Base-FX
 - 100Base-T4
- Switched Ethernet eliminates collisions using full-duplex links.



Key Point

Fast Ethernet achieved a tenfold increase in speed while preserving Ethernet compatibility.

Gigabit Ethernet (1000 Mbps)

- Developed to meet the demand for higher-speed LANs.
- Transmission rate increased to:

$$1000 \text{ Mbps} = 1 \text{ Gbps}$$

- Standardized as:



Design Goals

- Increase speed to 1 Gbps.
- Keep Ethernet addressing unchanged.
- Preserve frame format.
- Retain minimum and maximum frame sizes.

Objectives of Gigabit Ethernet

Gigabit Ethernet was designed to maintain compatibility with previous Ethernet versions.

Feature	Status
Address Length	Unchanged
Frame Format	Unchanged
Minimum Frame Size	64 Bytes
Maximum Frame Size	1518 Bytes
Data Rate	Increased to 1 Gbps

MAC Sublayer in Gigabit Ethernet

A major goal was to preserve the MAC sublayer.

However, achieving:

1 Gbps

required modifications to medium access methods.

Gigabit Ethernet supports:

- Half-Duplex Mode
- Full-Duplex Mode

Most modern networks use Full-Duplex mode.

Half-Duplex vs Full-Duplex

Half-Duplex	Full-Duplex
One direction at a time	Simultaneous send and receive
May experience collisions	No collisions
Requires CSMA/CD	No CSMA/CD
Rarely used today	Widely used today

Full-Duplex Gigabit Ethernet

In Full-Duplex operation:

- Each host is connected to a central switch.
- Dedicated point-to-point links are used.
- Data can be transmitted and received simultaneously.
- No shared medium exists.

Important

Since there is no shared medium, collisions cannot occur.

Role of the Switch

The switch performs the following functions:

- 1 Receives incoming frames.
 - 2 Stores frames in buffers.
 - 3 Reads destination MAC address.
 - 4 Forwards frame through the correct port.
 - 5 Delivers frame to destination.
- 

Result

Efficient frame forwarding without collisions.

Why CSMA/CD is Not Needed

Traditional Ethernet required:

CSMA/CD

because many stations shared the same medium.

Gigabit Ethernet uses:

- Switched topology
- Full-duplex communication
- Dedicated links

Therefore:

No Collision $\Rightarrow$ No CSMA/CD

Impact on Cable Length

In Standard Ethernet:

- Cable length was limited by collision detection.

In Gigabit Ethernet:

- No collision detection is required.
- Cable length depends mainly on:
 - Signal attenuation
 - Physical medium characteristics

Advantage

Supports longer distances without collision constraints.

Gigabit Ethernet Architecture

Host ↔ Switch ↔ Host

Characteristics:

- Dedicated point-to-point connections.
- Buffered switching.
- Full-duplex communication.
- No collisions.
- High throughput.

Gigabit Ethernet Implementations

Gigabit Ethernet supports different physical media.

Common implementation families:

- 1000Base-SX (Short-Wave Fiber)
- 1000Base-LX (Long-Wave Fiber)
- 1000Base-CX (Copper Cable)
- 1000Base-T (Twisted Pair)

S-W = Short Wave

L-W = Long Wave

Gigabit Ethernet Media Types

Implementation	Medium
1000Base-SX	Short-Wavelength Fiber
1000Base-LX	Long-Wavelength Fiber
1000Base-CX	Shielded Copper Cable
1000Base-T	Twisted Pair Cable

Fast Ethernet vs Gigabit Ethernet

Feature	Fast Ethernet	Gigabit Ethernet
Speed	100 Mbps	1000 Mbps
MAC Layer	Mostly Same	Modified
CSMA/CD	Optional	Rarely Used
Switching	Common	Essential
Full Duplex	Supported	Standard

Advantages of Gigabit Ethernet

- 10 times faster than Fast Ethernet.
- High throughput.
- Collision-free communication.
- Full-duplex operation.
- Compatible with existing Ethernet standards.
- Supports fiber and copper media.
- Ideal for backbone networks and data centers.

Summary of Gigabit Ethernet

- Gigabit Ethernet operates at:

1 Gbps

- Standardized as IEEE 802.3z.
- Preserves Ethernet frame format and addressing.
- Uses mainly full-duplex communication.
- Eliminates collisions and CSMA/CD.
- Relies on switches and dedicated links.
- Supports fiber-optic and copper implementations.

Key Point

Gigabit Ethernet achieves very high speed through full-duplex switched communication while remaining compatible with Ethernet standards.

10-Gigabit Ethernet (10 Gbps)

- Designed to extend Ethernet beyond traditional LANs.
- Supports:
 - Local Area Networks (LANs)
 - Metropolitan Area Networks (MANs)
 - Wide Area Networks (WANs)
- Standardized as:

IEEE 802.3ae

- Data rate:

10 Gbps

Goals of 10-Gigabit Ethernet

The primary design goals were:

- ① Increase data rate to:
10 Gbps
- ② Retain Ethernet frame format.
- ③ Preserve minimum and maximum frame sizes.
- ④ Enable Ethernet operation across:
 - LANs
 - MANs
 - WANs
- ⑤ Provide high-speed backbone connectivity.

Evolution of Ethernet Speeds

Ethernet Version	Data Rate
Standard Ethernet	10 Mbps
Fast Ethernet	100 Mbps
Gigabit Ethernet	1 Gbps
10-Gigabit Ethernet	10 Gbps

Observation

Each generation increased bandwidth by approximately 10 times.

10-Gigabit Ethernet currently relies on:

Fiber-Optic Communication

Reasons:

- Extremely high bandwidth
- Low attenuation
- Long-distance communication
- Better signal quality

Physical Layer Types

10-Gigabit Ethernet defines two physical-layer variants.

Physical Layer	Purpose
LAN PHY	Supports existing LAN infrastructures
WAN PHY	Supports WAN communication using SONET OC-192

LAN PHY vs WAN PHY

LAN PHY	WAN PHY
Used inside LANs	Used across WANs
High-speed Ethernet backbone	Interconnects distant networks
Ethernet-based	SONET OC-192 compatible

Medium Access Method

10-Gigabit Ethernet operates only in:

Full-Duplex Mode

Characteristics:

- Simultaneous transmission and reception.
- Dedicated point-to-point links.
- No shared medium.
- No collisions.

Why CSMA/CD is Not Required

Traditional Ethernet required:

CSMA/CD

because stations shared the same medium.

10-Gigabit Ethernet:

- Uses full-duplex communication.
- Employs switched networks.
- Eliminates collisions completely.

No Collision $\Rightarrow$ No CSMA/CD

Architecture of 10-Gigabit Ethernet

Host ↔ Switch ↔ Host

Features:

- Dedicated links.
- Buffered switching.
- Collision-free operation.
- High throughput.
- Long-distance connectivity.



Common Implementations

The most common implementations are:

- 10GBase-SR
- 10GBase-LR
- 10GBase-EW
- 10GBase-X4



Refer to Table 5.9 for detailed specifications.

10-Gigabit Ethernet Implementations

Implementation	Description
10GBase-SR	Short-Range Fiber
10GBase-LR	Long-Range Fiber
10GBase-EW	WAN-Compatible Ethernet
10GBase-X4	Four-Lane High-Speed Ethernet

Gigabit vs 10-Gigabit Ethernet

Feature	Gigabit	10-Gigabit
Speed	1 Gbps	10 Gbps
Full Duplex	Yes	Yes
CSMA/CD	Rarely Used	Not Used
Fiber Support	Yes	Yes
LAN Support	Yes	Yes
WAN Support	Limited	Yes

Advantages of 10-Gigabit Ethernet

- Extremely high data rate.
- Suitable for backbone networks.
- Supports LAN, MAN, and WAN integration.
- Full-duplex communication.
- No collisions.
- Compatible with Ethernet standards.
- Scalable for future networking needs.

Summary of 10-Gigabit Ethernet

- Operates at:

10 Gbps

- Standardized as IEEE 802.3ae.
- Uses fiber-optic technology.
- Supports LAN PHY and WAN PHY.
- Works only in full-duplex mode.
- Eliminates collisions and CSMA/CD.
- Enables Ethernet deployment across LANs, MANs, and WANs.

Key Point

10-Gigabit Ethernet extends Ethernet technology beyond local networks while maintaining compatibility with existing Ethernet standards.